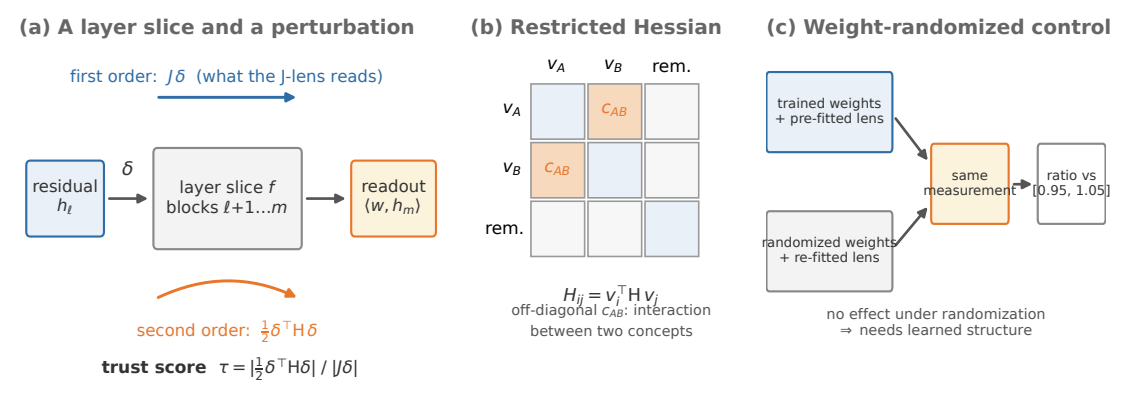
(a) A layer slice and a perturbation
first order: $J\delta$ (what the J-lens reads)
residual h_ℓ
δ
layer slice f blocks $\ell{+}1 \ldots m$
readout $\langle w, h_m \rangle$
second order: $\frac{1}{2}\delta^\top \mathrm{H}\, \delta$
trust score $\tau = |\frac{1}{2}\delta^\top \mathrm{H}\delta| \,/\, |J\delta|$
(b) Restricted Hessian
v_A
v_B
rem.
c_{AB}
c_{AB}
$H_{ij} = v_i^\top \mathrm{H}\, v_j$
off-diagonal c_{AB}: interaction between two concepts
(c) Weight-randomized control
trained weights + pre-fitted lens
randomized weights + re-fitted lens
same measurement
ratio vs [0.95, 1.05]
no effect under randomization
⇒ needs learned structure